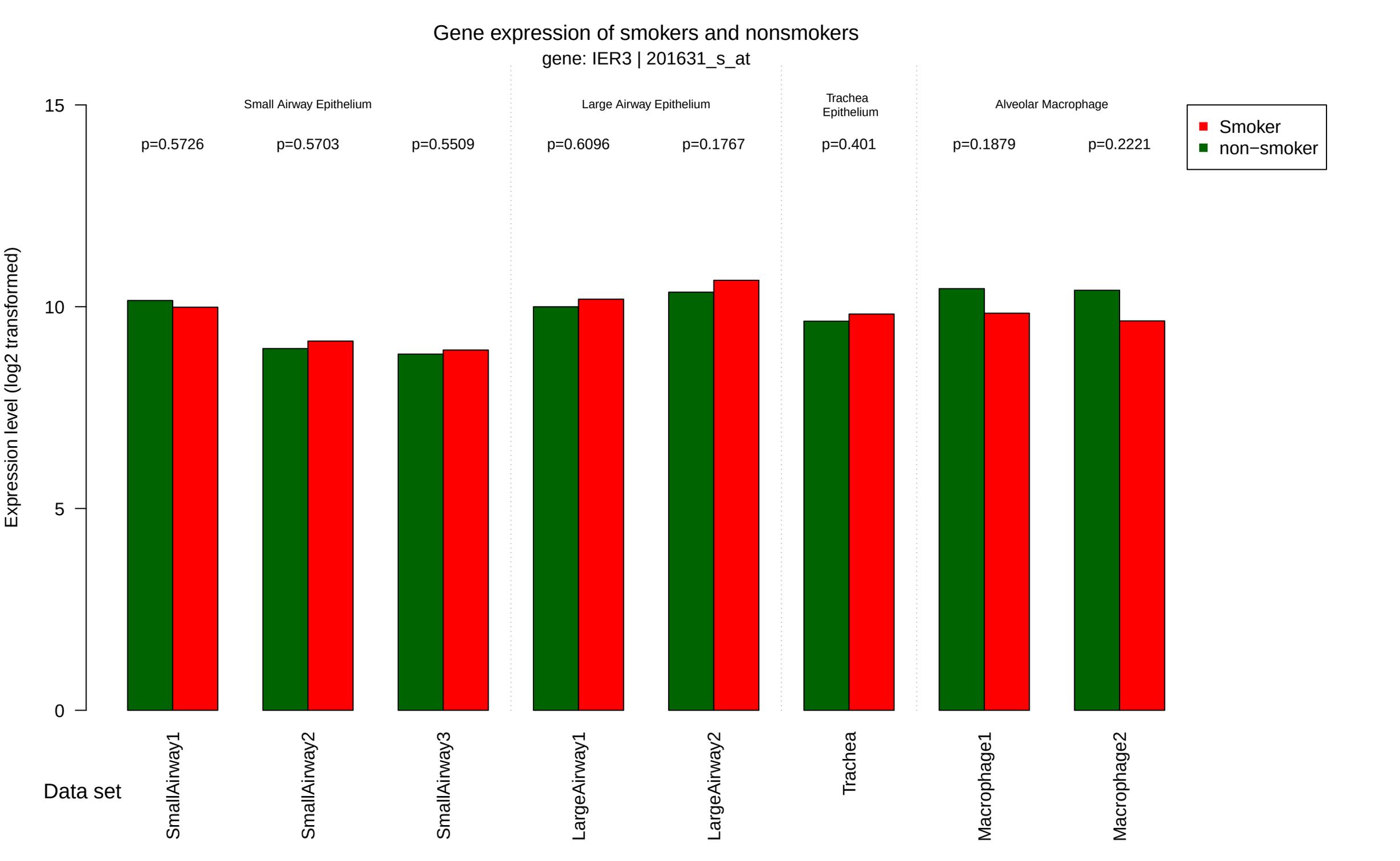
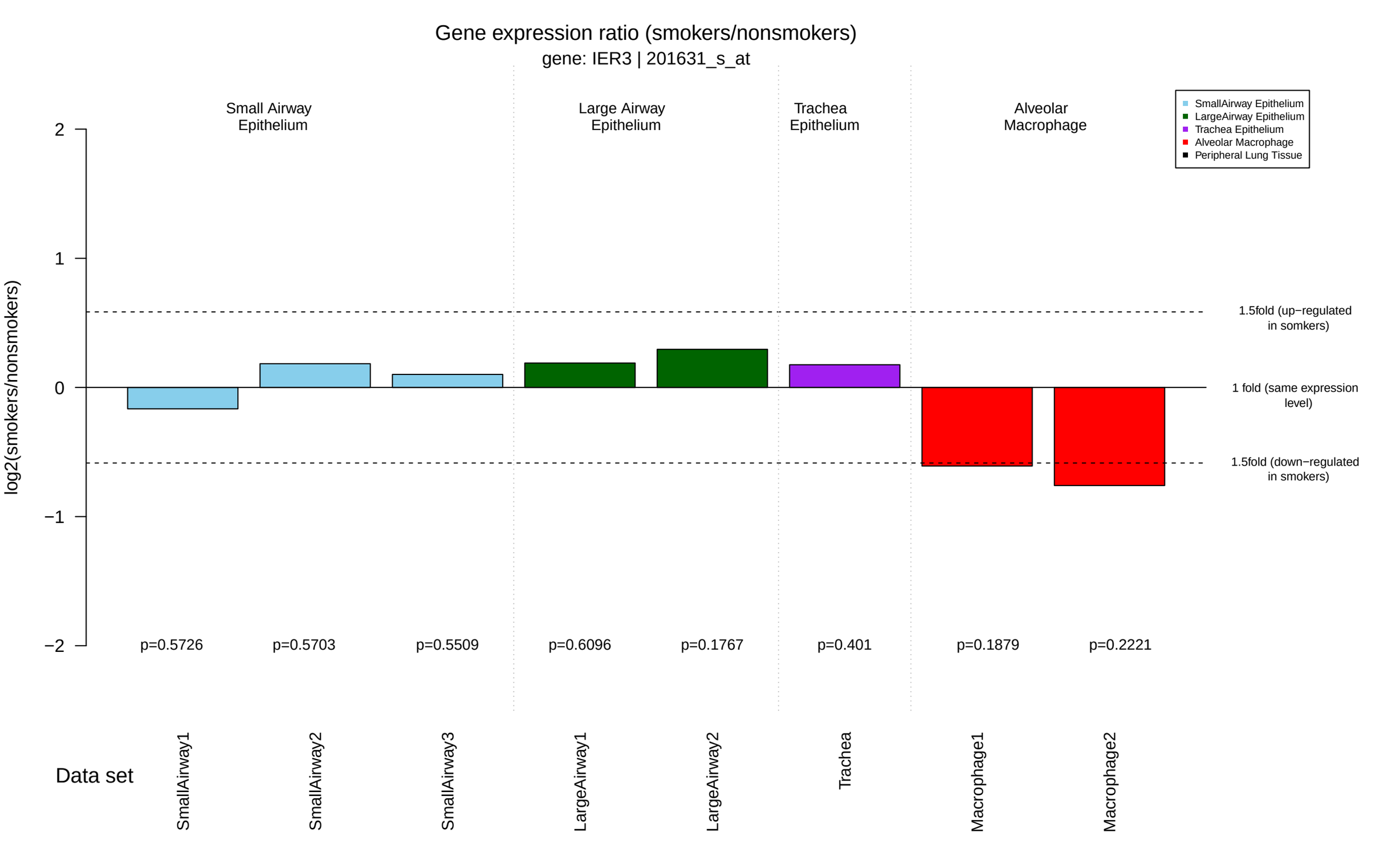
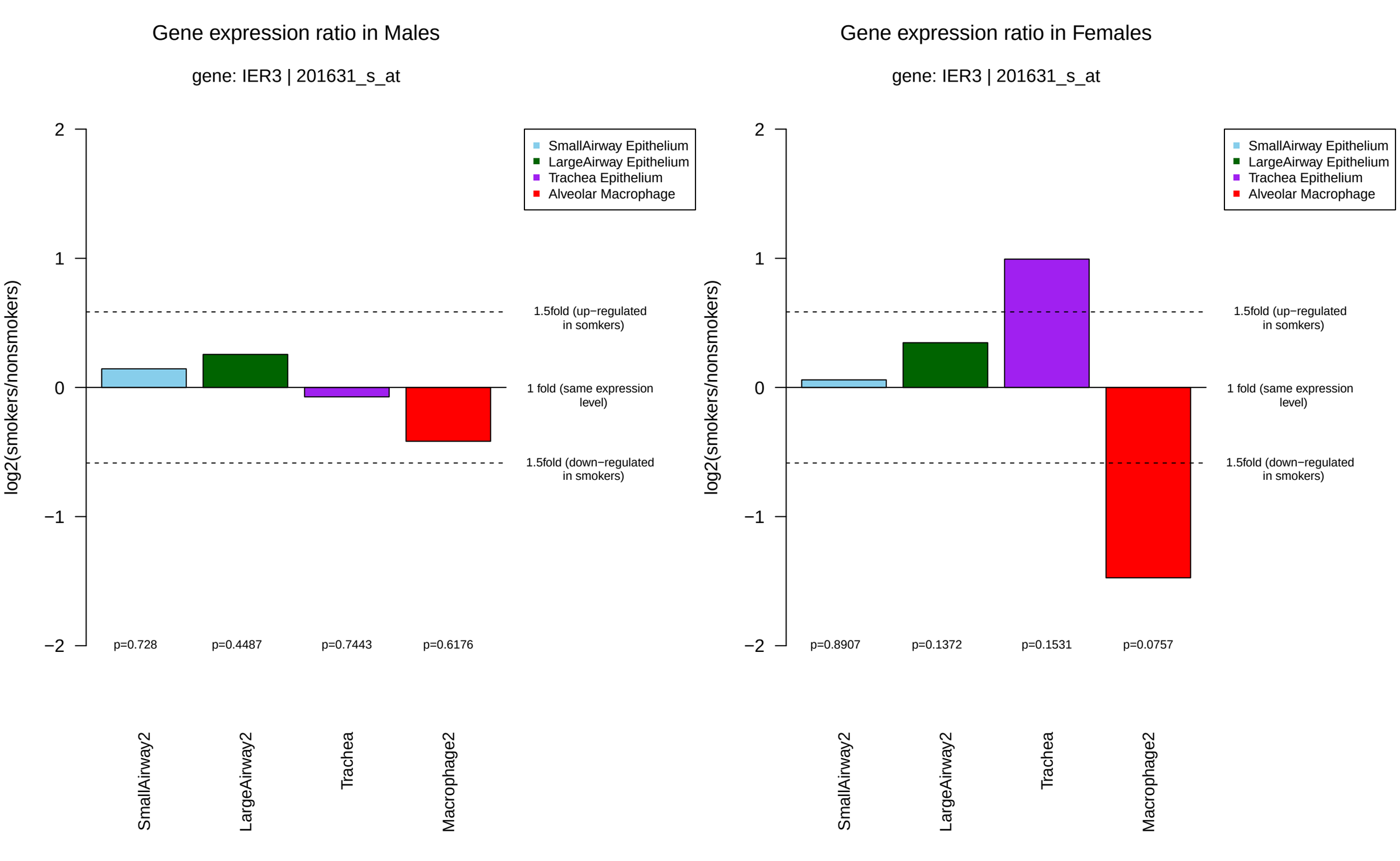
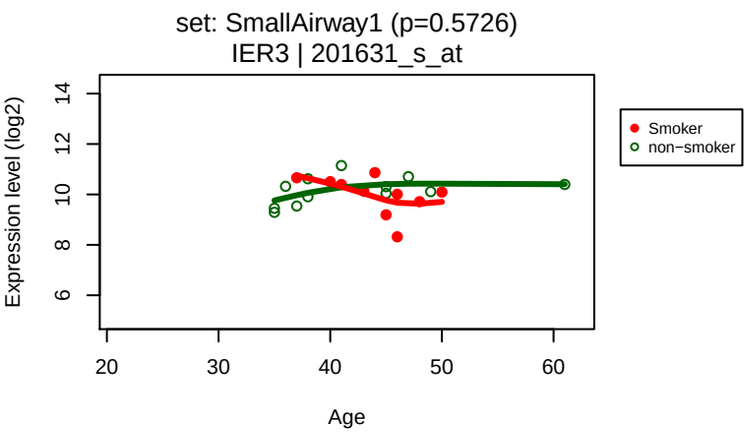
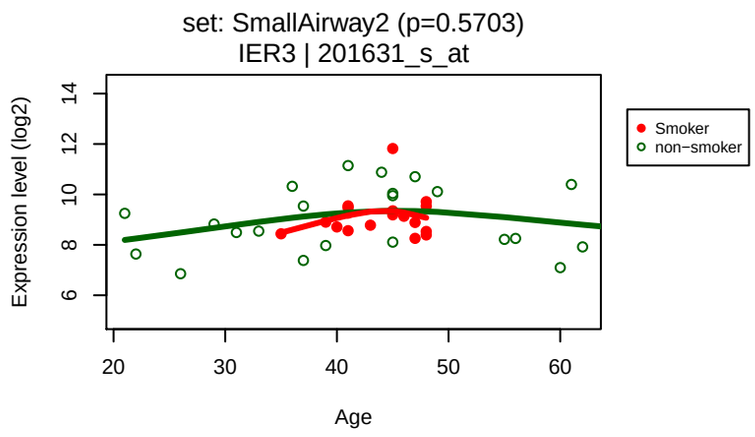
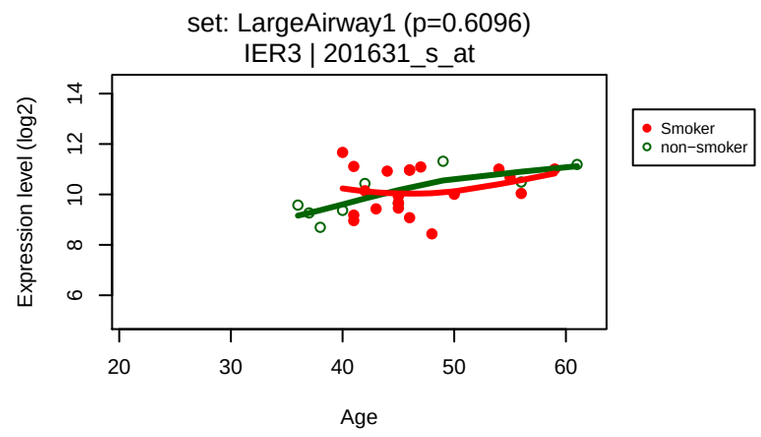

Gene expression of smokers and nonsmokers
gene: IER3 | 201631_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5726
p=0.5703
p=0.5509
p=0.6096
p=0.1767
p=0.401
p=0.1879
p=0.2221
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

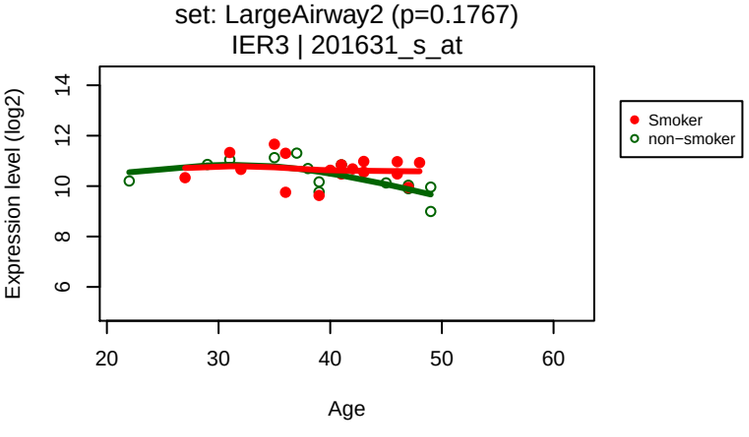

Gene expression ratio (smokers/nonsmokers)
gene: IER3 | 201631_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5726
p=0.5703
p=0.5509
p=0.6096
p=0.1767
p=0.401
p=0.1879
p=0.2221
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

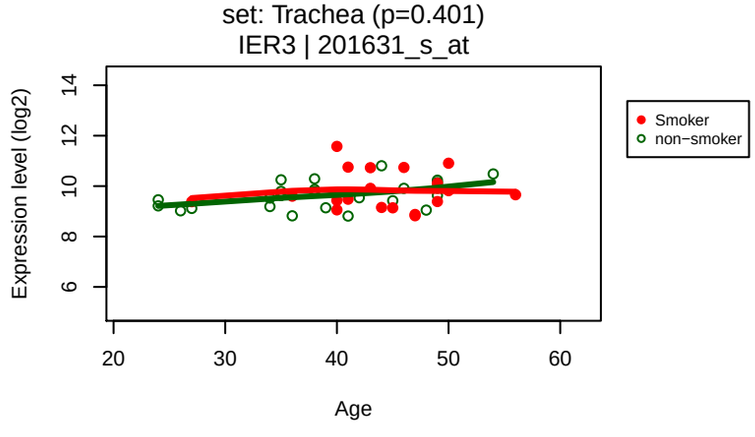

Gene expression ratio in Males
gene: IER3 | 201631_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.728
p=0.4487
p=0.7443
p=0.6176
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IER3 | 201631_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8907
p=0.1372
p=0.1531
p=0.0757
SmallAirway2
LargeAirway2
Trachea
Macrophage2

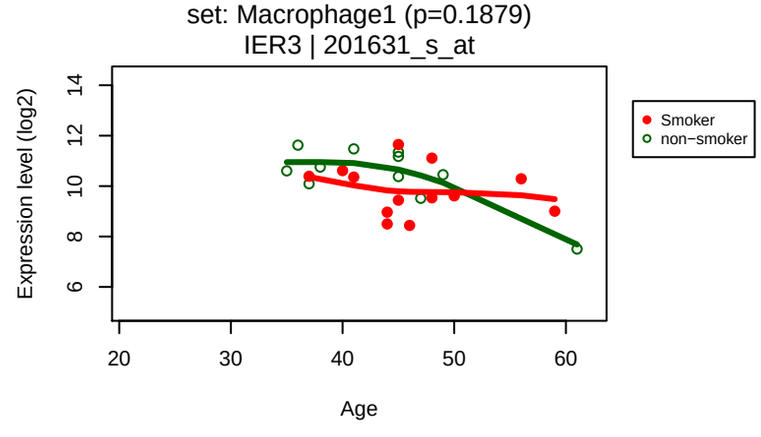

set: SmallAirway1 (p=0.5726)
IER3 | 201631_s_at
Expression level (log2)
Age
Smoker
non−smoker

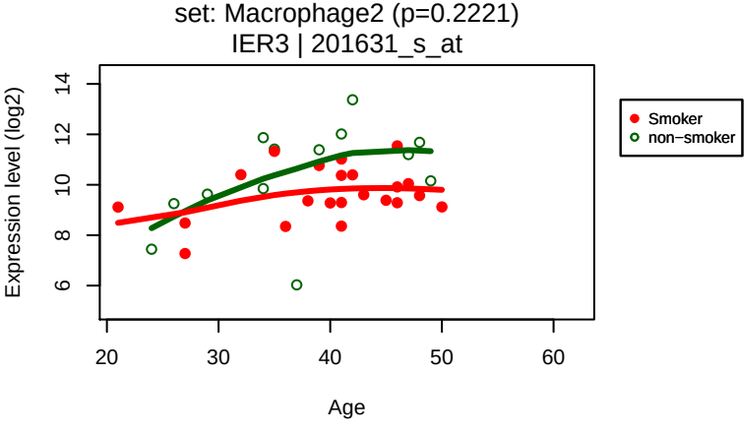

set: SmallAirway2 (p=0.5703)
IER3 | 201631_s_at
Expression level (log2)
Age
Smoker
non−smoker

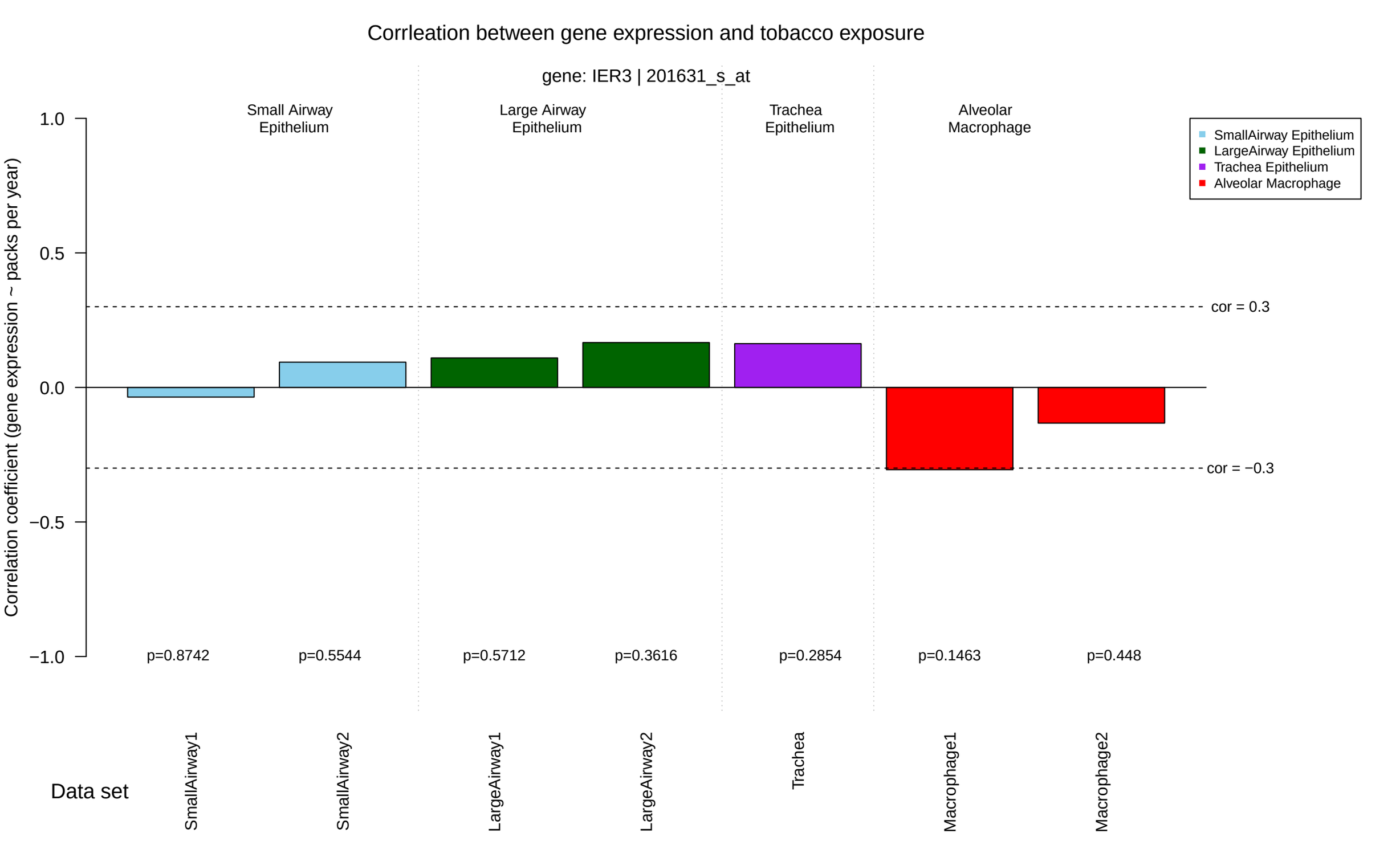

set: LargeAirway1 (p=0.6096)
IER3 | 201631_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1767)
IER3 | 201631_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.401)
IER3 | 201631_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1879)
IER3 | 201631_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2221)
IER3 | 201631_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IER3 | 201631_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8742
p=0.5544
p=0.5712
p=0.3616
p=0.2854
p=0.1463
p=0.448
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2